Revenue Horizon Forecasting

A reproducible 90-day ARIMA planning workflow

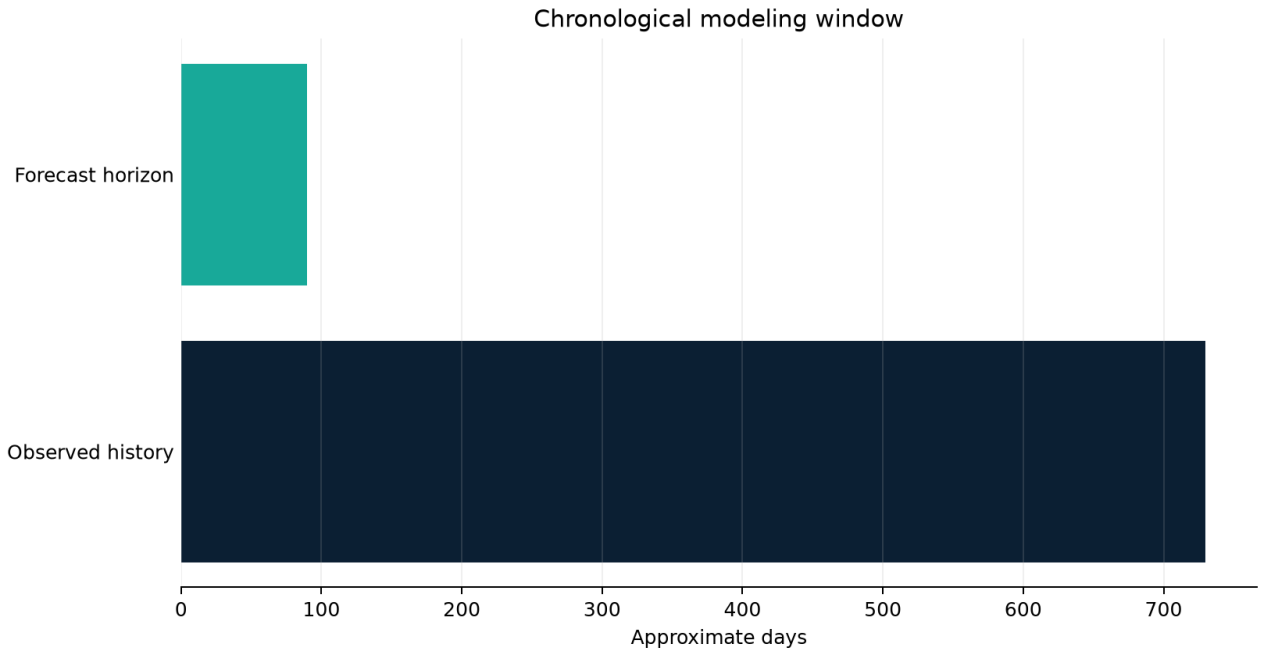
Portfolio Case Study

Executive Summary

This time-series workflow transforms daily revenue history into a 90-day forecast with uncertainty intervals and holdout evaluation. The source analysis found a non-stationary revenue series, achieved stationarity after first differencing, and selected an ARIMA(1,1,0) specification. The public implementation emphasizes temporal ordering, explicit validation, and forecast intervals rather than point estimates alone.

Decision Question

How can two years of daily revenue history support a decision-ready 90-day planning forecast?



Methods

- Validated date continuity, ordering, numeric revenue values, and minimum sample size.
- Diagnosed trend and stationarity; first differencing produced an ADF p-value below 0.01 in the source analysis.
- Used chronological holdout evaluation rather than random splitting.
- Generated 90 daily forecasts with 95% uncertainty intervals and export-ready tabular output.

Evidence and Findings

Evidence	Observed result
Historical window	Approximately two years of daily observations support the planning horizon.
Stationarity	The raw series was non-stationary; first differencing produced a stationary modeling series.
Model	ARIMA(1,1,0) was selected in the source analysis as a compact fit for trend-adjusted behavior.
Decision horizon	Ninety forecast days align the output to near-term capacity and resource planning.

Limitations

- A univariate model cannot explain promotions, pricing changes, outages, or other external drivers.
- Forecast intervals reflect model uncertainty, not every operational shock.
- Model order and error should be re-evaluated as new observations arrive.

Recommended Next Actions

- Refresh the forecast on a fixed cadence and track holdout MAE and RMSE over time.
- Use interval-aware planning scenarios instead of a single deterministic revenue value.
- Add validated external regressors only when they are available at forecast time.

Reproducibility

The companion repository contains typed Python modules, tests, dependency declarations, data-access notes, and a rendered analysis notebook. Restricted source data and the working notebook are intentionally excluded from the public repository.

Results are reported as historical analytical evidence. They should be reproduced on an authorized equivalent dataset before operational use.